ACC3018 Applied Analytics Capstone

Seminar 1

Agenda

- Module Briefing

- Essentials of Research (to learn)
  - Key issues in research
  - Thesis structure

- In-Class-Activity (to apply on a real research example)
  - In-Class-Activity 1: Literature Review
  - In-Class-Activity 2: Hypotheses Development
  - In-Class-Activity 3: Research Design

Module Briefing

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AY2024/25 T3 (15%)	AY2025/26 T1 (35%)	AY2025/26 T2 (50%)

• Research Methods (6 Seminars)
  • S1: Essentials of Applied Research
  • S2-S3: Research Methods
  • S4-S5: Data Application
• Self-Study
  • IRB
  • SIT Library Search
  • How to write Literature Review
  • How to write research proposal
  • How to write final report
	• Conduct Capstone Project
	  • Submit and Present Research Proposal
	  • Consultation with Academic Supervisor and CPC team
	  • During BTC workshops
		• Conduct and Complete Capstone Project
		  • Submit Research Final Report
		  • Consultation with Academic Supervisor and CPC team during BTC workshops

IWSP

Module Briefing - Assessment Tasks

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AY2024/25 T3 (15%)

- In-class-Activity and Attendance (5%) – Individual S1-S5
- S6: Mini Group Assignment (10%)

AY2025/26 T1 (35%)

- Research Proposal (35%)
  - Introduction
  - Literature Review and Hypotheses Development
  - Research Design

AY2025/26 T2 (50%)

- Research Final Report (50%)
  - Introduction
  - Literature Review and Hypotheses Development
  - Research Design
  - Results (incl. Robustness check)
  - Conclusion

Group Work (5-6 students per group), Peer Review (Sparkplus)

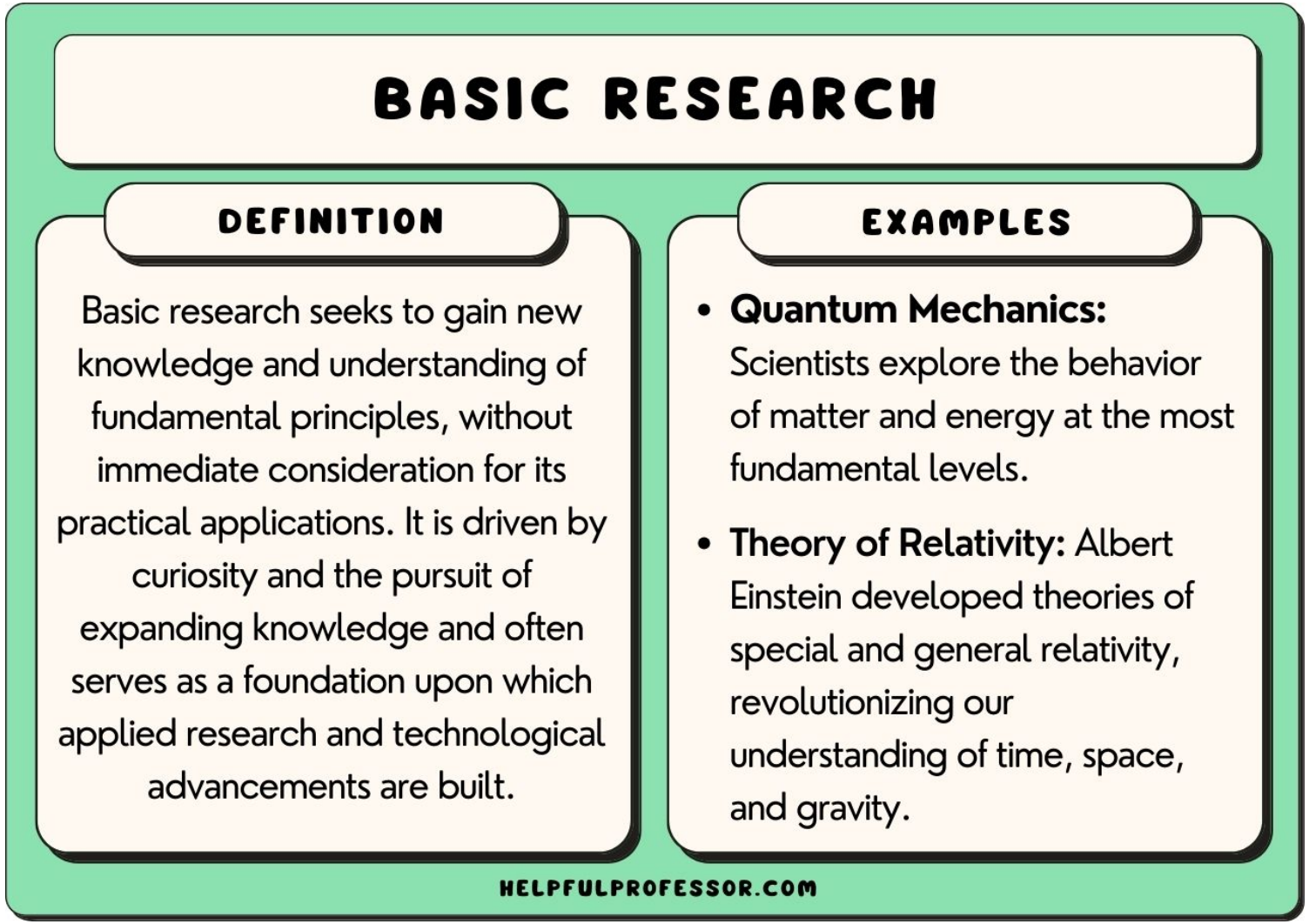
Essentials of Research – What is Research?

Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon with which we are concerned or interested.

Grinnell (1993):"Research is a structured inquiry that utilizes acceptable scientific methodology to solve problems and creates new knowledge that is generally applicable"

Burns (1997) defines research as "a systematic investigation to find answers to a problem"

Basic Research

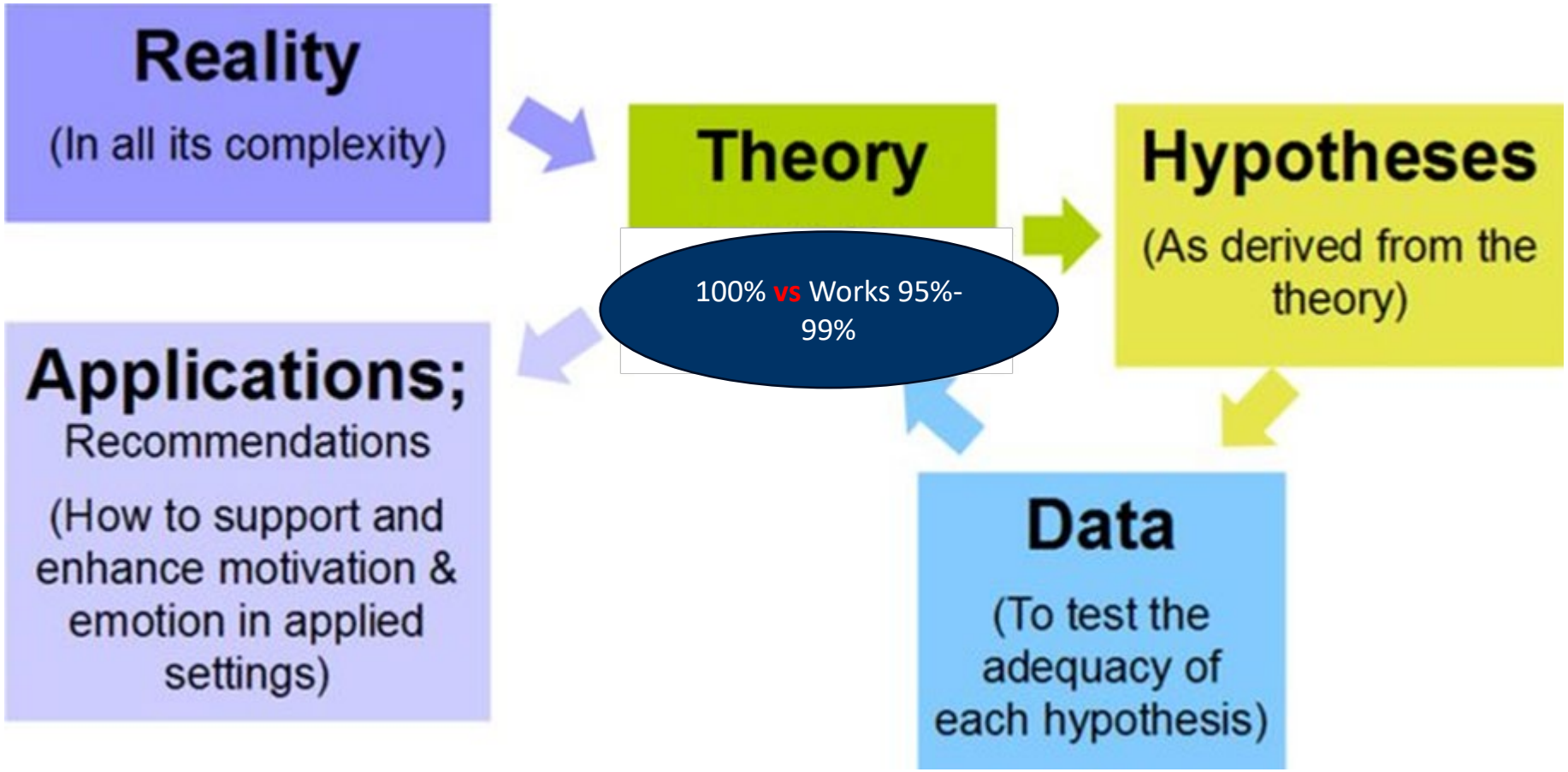


- **More Examples of basic research**
  - Investigation into how the universe began.
  - A study searching for the causes of cancer.
  - Understanding the components that make up human DNA.
  - An examination into whether a vegetarian diet is healthier than one with meat.
  - A study to learn more about which areas in the world get the most precipitation.

Basic VS Applied Research

Parameters	Basic Research	Applied Research
Type of knowledge produced	Scientific Discovery (Science)	Technological Application (Technology)
Motivation	Intellectual curiosity	Solving problems
Key questions	Is it true ?	Does it work ?
Objective	To understand	To come up with solutions

Basic VS Applied Research



The Research Process

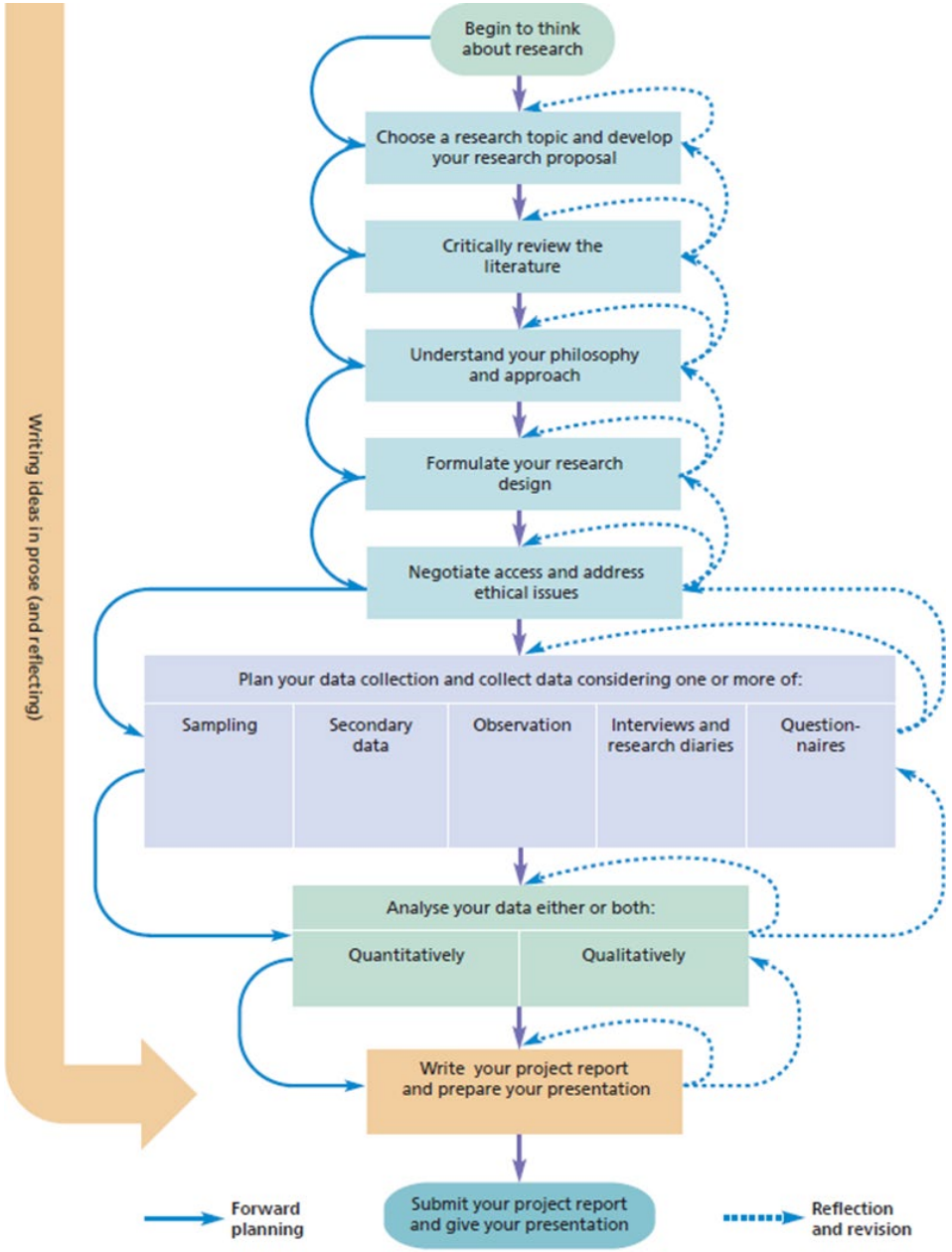


Figure 1.2 The research process

Source: © Mark Saunders, Philip Lewis and Adrian Thornhill 2018

How to Generate Research Ideas?

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Table 2.1 More frequently used techniques for generating and refining research ideas

Rational thinking	Creative thinking
Examining your own strengths and interests	Keeping a notebook of your ideas
Examining staff research interests	Exploring personal preferences using past projects
Looking at past project titles	Exploring relevance to business using the literature
Discussion	Relevance trees
Searching existing literature	Brainstorming
Scanning the media	

Brainstorming

Financial Accounting
- Earnings management
- Fair value vs. historical cost accounting
- Financial statement analysis and forecasting
- Accounting conservatism
- Impact of financial reporting standards

Managerial (Management) Accounting
- Performance measurement and incentives
- Cost behavior and cost management
- Budgeting, planning, and control systems
- Strategic management accounting
- Sustainability and ESG reporting

Auditing
- Audit quality and auditor independence
- Audit market structure and competition
- Fraud detection and forensic accounting
- Regulation and oversight

Taxation
- Tax avoidance and tax planning
- Impact of tax policy on firm behavior
- International taxation and transfer pricing
- Behavioral responses to taxation

Finance
- Asset Pricing and Risk Management
- Corporate Finance
- Banking and Financial Intermediation
- Sustainable and ESG Finance
- Behavioural Finance
- Household and personal finance

Accounting Information Systems (AIS)
- ERP systems and accounting practices
- Blockchain and digital ledger technology in accounting
- Data analytics and artificial intelligence in auditing

Characteristics Of A Good Research Topic

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Box 2.1 Checklist

Characteristics of a good research topic

Appropriateness

- ✔ Does the research topic fit the specifications and meet the standards set by the examining institution?
- ✔ Does the research topic contain issues that have a clear link to theory?
- ✔ Are you able to state your research question(s), aim and objectives clearly?
- ✔ Will the proposed research be able to provide fresh insights into this topic?
- ✔ Are the findings for this research topic likely to be symmetrical: that is, of similar value whatever the outcome?

Capability

- ✔ Do you have, or can you develop within the project time frame, the necessary research skills to undertake the research topic?
- ✔ Is the research topic achievable within the available time?
- ✔ Is the research topic achievable within the financial resources that are likely to be available?
- ✔ Are you reasonably certain of being able to gain access to data you are likely to require for this research topic?

Fulfilment

- ✔ Does the research topic really interest and motivate you?
- ✔ Will the research topic help towards the achievement of your future aspirations or career goals?

How Do I Choose A Research Topic?

Table 1.2 Practitioner and management researcher orientations

Management researcher		Practitioner
Basic understanding	**Focus of interest**	Usable knowledge
General enlightenment		Instrumental
Theoretical explanation		Practical problem solutions
'Why' knowledge		'How to' knowledge
Substantive theory building		Local theory-in-use
Theoretical and methodological rigour	**Methodological imperative**	Timeliness
Academic publication	**Key outcome**	Actionable results with practice impact
Disdain of practitioner	**Views of other**	Deprecate or ignore
Desire to make a difference to practice		Belief research can provide relevant (socially useful) fresh insights to managers' problems

Source: Developed from Saunders (2011)

Examples of Past Year ACC Capstone Projects

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- The perceived benefits and challenges of KAMs disclosures – Stakeholders' perspectives in Singapore
- Impact of Audit Quality on the Financial Position of Listed Companies'
- Exploring Expectations & Perspectives on the Skills and Competencies Required for a Successful Career in Audit
- **How does a firm's practices affect the attraction and retention of talents in the accounting profession?**
- How has cryptocurrencies affect luxury goods?
- To study the carbon footprint of the publishers for SIT Library to evaluate the extent of "greenness" of its suppliers and refine its procurement procedures
- How does public sentiment of a company's ESG affect its financials?
- **How do the Environmental and Social components of ESG affect companies' operating performance across various industries?**
- Assessing students' perceptions and awareness of sustainability issues at Singapore higher education institutions
- Impact of Carbon Tax on Greenhouse Gas Emissions in Singapore
- Does audited sustainability reports have a positive relationship with financial performance?
- The exploration of Board Composition and ESG Initiatives
- Using ChatGPT to help processing MD&A information
- **Evaluating ChatGPT's capabilities in CA Qualification Exams**
- Using ChatGPT for ESG Reporting
- Can we use machine learning models to reliably predict corporate bankruptcy?
- Business Process Reengineering (BPR): Determining Service Workflow & Solutions for Remote Touchpoints for Singapore Institute of Technology (SIT) Punggol Library

Why Do We Need A Literature Review?

- To conduct a 'preliminary' search of existing material to help to generate and refine your research ideas and draft your research proposal
- To develop a critical perspective – provides the context and theoretical framework for your research
- To identify other research that may be in progress and to place your research findings within the wider body of knowledge and forms part of your discussion chapter
- It can tell you if a research question has already been answered.
- It can help you evaluate the interestingness of a research question.
- It can give you ideas for how to conduct your own study.
- It can tell you how your study fits into the research literature

Literature Review Examples – Earnings Quality

No	Attributes	Measurements	Researcher
1	Accrual Quality	*Mapping* the current accrual towards the past, present and future cash flow	Dechow and Dichev (2002); Largay III (2002); Francis et al. (2003a, 2003b, 2004); Chambers (2003)
2	Persistance	The regression coefficient of current earnings towards the future earnings (Model AR1)	Francis et al. (2003b, 2004), Sloan (1996)
3	Predictability	The standard deviation of model AR1 error	Lougee and Marquardt (2002); Francis et al. (2003a); Brown and Sivakumar (2001)
4	Smoothness	The ratio of earnings variability towards the cash flow variability.	Francis et al. (2003b, 2004)

Literature Sources

1. Refereed (peer-reviewed) academic journal
2. Non-refereed academic journal
3. Professional journal
4. Trade journals/magazines
5. Books and e-books
6. Newspapers
7. Conference proceedings
8. Reports
9. Theses

Top Journals

Discipline	No. of articles	Average no. of pages per article	SNIP impact factor
Accounting (4 journals)	1765	25.06	2.53
Accounting Review	596	27.66	2.31
Accounting, Organizations and Society	410	19.18	2.66
Journal of Accounting Research	371	33.23	2.28
Journal of Accounting and Economics	388	20.16	2.87
Economics (6 journals)	4224	32.09	4.42
American Economic Review	1237	23.20	3.45
Annals of Statistics	1062	28.19	2.92
Econometrica	625	32.38	4.72
Journal of Political Economy	346	36.97	4.89
Quarterly Journal of Economics	438	42.97	6.86
Review of Economic Studies	516	28.86	3.70
Finance (3 journals)	2867	31.69	4.13
Journal of Finance	792	35.35	4.83
Journal of Financial Economics	1182	23.12	3.92
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Why Do We Need A Literature Review?

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Answers to these questions will usually help define a specific hypothesis or research question.

Q. What types of research have been done in the area?

Q. What has been found in previous studies?

Q. What suggestions do other researchers make for further study?

Q. What has not been investigated?

→ The gap in the research area, to identify your research question → Develop your hypotheses

Q. How can the proposed study add to our knowledge of the area?

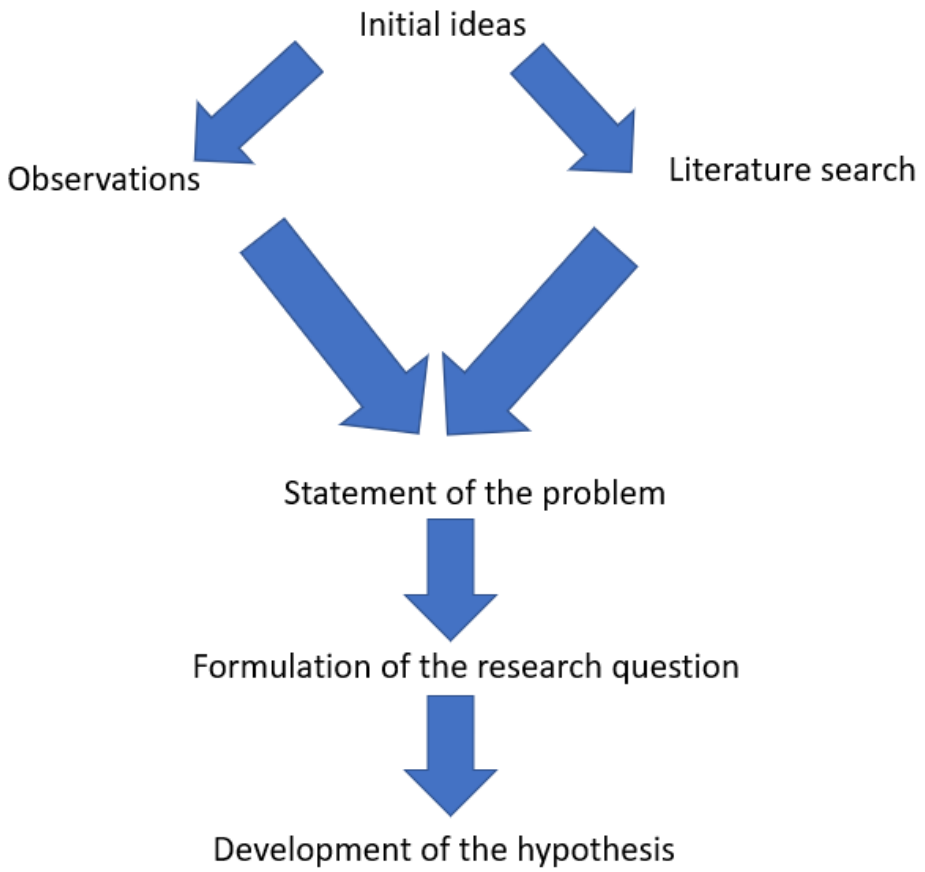
→ The contribution of your study

Q. What research methods were used in previous studies?

→ Develop your research design

Literature Review and Hypotheses Development

- A thoroughly researched literature review is expected to contain sub-sections that include the following :
  - theoretical basis to the RQ;
  - **relevant** findings of past studies;
  - how **EACH** hypothesis is developed;
    - establish linkage between topics
  - state the hypothesis.



How to Formulate Hypotheses?

Make a prediction based on previous observations or research. (**Provide theoretical argument, reasoning**, etc.) **e.g. ESG improve firm performance. Why?**

Define objective independent (**X e.g. ESG**) and dependent variables (**Y e.g. Performance**).

Make sure the hypothesis is testable (**to develop regression models**) and falsifiable (**Null and Alternative Hypotheses**).

Hypotheses Development

- When **formulating** the hypothesis:
  - express a relation between X and Y variables.
  - wording should be clear and unambiguous
  - imply possibilities for testing **(from research design)**
    - type of dataset (cross-sectional, time-series or panel)
    - unit of observation (firm, individual, etc.)
    - nature of the variables
      - Continuous (e.g. ROA), dummy (e.g. gender), index etc.
      - independent, dependent etc.
    - nature of the research methodology
      - Linear or non-linear

Hypotheses Development

- When **stating** the hypothesis, **either** state both the null (**H_0**) and alternate hypotheses (**H_a**) **OR** just the alternate hypothesis (**H_a**)

Example:

Hypothesis 1. (H_a): Firms with higher (lower) ESG scores performed better (poorer), ceteris paribus.

→ Expect a positive relationship between X and Y in regression analysis

→ ceteris paribus: all else equal, which include other Xs that statistically significant in explaining Y in the literature

Research Design

- Develop Regression Models to test the hypotheses;
- Explain the key variables and control variables;
  - State variable name, type, definition and source
  - Include all variable definitions in the Appendix
- Based on the hypothesis formulated, what is your expected coefficient estimate?
  - E.g., Based on H1, this study expects a positive and **statistically**
    - **significant** coefficient estimate for….

Research Design

Example (Y=a+**b**X1+Cx2+dX3+e, clearly defined X1 to explain Y)**:**

- According to the capital asset pricing model (CAPM), the expected return on a stock, ***E(R$_i$),*** is the sum of the risk-free rate of interest, ***R$_F$***, and a premium for bearing the stock's market risk, **β$_i$**(***R$_M$ − R$_F$***):

$$E(R_i) = R_F + \beta_i\left[E(R_M) - R_F\right]$$

β_i = the return sensitivity of stock i to changes in the market return

$E(R_M)$ = the expected return on the market

$E(R_M) - R_F$ = the expected market risk premium

- According to the Fama French 3-factor model:

$$R_{it} - R_{ft} = \propto_{it} + \beta_1(R_{mt} - R_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \varepsilon_{it}$$

- The contributions are that SMB and HML are significant (**β$_2$** , **β$_3$**) in explaining excess return, after controlling for expected market risks.

Research Design

- Data and Sample Selection
  - Data sources i.e. What from where? (Data Set, Time Horizon)
  - Criteria for selection
  - How is the final sample constructed? (Sample size)
- Model specifications (Regression Models, Variable definitions)
- Use of precise notations e.g., i, t…

Results

- Summary statistics and/or Data Analytics (Descriptives, Correlation Matrix)
- Baseline/Main results (Interpret, Evidence support/against hypotheses? Implication?)
- Robustness tests (Multicollinearity? Endogeneity?)
- Note:
  - All tables should be **standalone**.
  - Analyses that are not central to the RQ should be included in the Appendix.

Conclusion

- **Reiterate** what your study is about
- What you have done to address the RQ
- Did you find support for each hypothesis?
  - If yes, why?
  - If no, why not?
- **Highlight the practical implications from the results in your study.**
- Suggest limitations and/or avenues for future research

References and Appendices

- **ONE** reference list:
  - using APA style
  - arranged in alphabetical order
  - according to the last name of the first author
- All additional analyses in the Appendix should be supported with brief explanations of its rationale and result.

Theses Structure

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- Abstract
- Introduction
- Literature Review & Hypothesis Development
- Research Design
- Results
- Conclusion
- References
- Appendices

Abstract

- This is usually the **LAST** of all sections to be written.
- One paragraph to outline:
  - Research question (RQ)
  - Main finding(s)
  - Robustness of finding(s) through various tests
- A neatly written abstract should be short and crisp but still communicates clearly why the study is a worthy read.

Introduction

- This is usually the section that sees the **HIGHEST** number of revisions.

- This section communicates the scope of your study and outlines the basis of this research.

- A well-structured introduction is expected to contain the following:
  a) **Theoretical basis of your RQ** and/or the observed phenomenon that underpins the RQ

Introduction

A well-structured introduction is expected to contain the following:

b) Motivations for this study

- How a) leads to the RQ → **Industry relevance?**

Examples:

- A topic that is overlooked?
- Literature gap?
- Conflicting results from past studies?

→ Provide strong reasons to justify why this study is worth doing!

Introduction

A well-structured introduction is expected to contain the following:

c) Your **approach** to address the RQ – briefly mention data sources

d) What does your **main finding**(s) suggest in relation to a) and the RQ?

e) Alternative explanations that may bias your finding(s) and robustness tests.

f) Contributions

- How does your study add value to this area of research? **Novelty?**
- Relate to your motivations

Introduction

A well-structured introduction is expected to contain the following:

g) **Implications**

- So what? What are the key takeaway(s)?
- **Who would be interested in this study? Industry players? Why would they be interested? → Applied research**

h) Highlight any studies that are of a similar nature (**if applicable**)

- How is your study different from theirs?
- How might your study be more superior?

i) The last paragraph in the introduction guides readers to the relevant sections.

- The remainder of this study is organised as follows. The next section provides…Section 3…

Research Proposal (To be submit in AY2025/26 T1)

- Abstract
- Introduction
- Literature Review & Hypothesis Development
- Research Design
- Results
- Conclusion
- References
- Appendices

A research proposal maps out what is to be done ***without*** the actual data collection, findings, discussion and conclusion.

In Class Activity – Individual work

Find a paper different from my example“ Earnings Management Through Real Activities Manipulation” (Roychowdhury 2006 JAE) from the top journals:

- Type journal name in OneSearch, find the latest issue and select one paper (choose different paper from your team/friends) to download
- Cite the paper and copy the abstract of the paper, identify its keywords
- **In-class activity 1:** Literature review
  - Identify your areas of literature
  - Identify the paragraph where gap is identified
- **In-class activity 2**: Hypotheses development
  - Motivation of the hypotheses
  - Identify the hypotheses
- **In-class activity 3**: Research design
  - How to measure the key concept in the hypotheses
  - How to test the hypotheses
  - Any Limitations

Discipline	No. of articles	Average no. of pages per article	SNIP impact factor
Accounting (4 journals)	1765	25.06	2.53
Accounting Review	596	27.66	2.31
Accounting, Organizations and Society	410	19.18	2.66
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"Earnings Management Through Real Activities Manipulation" (Roychowdhury 2006 JAE)

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Abstract

I find evidence consistent with managers manipulating real activities to avoid reporting annual losses. Specifically, I find evidence suggesting price discounts to temporarily increase sales, overproduction to report lower cost of goods sold, and reduction of discretionary expenditures to improve reported margins. Cross-sectional analysis reveals that these activities are less prevalent in the presence of sophisticated investors. Other factors that influence real activities manipulation include industry membership, the stock of inventories and receivables, and incentives to meet zero earnings. There is also some, though less robust, evidence of real activities manipulation to meet annual analyst forecasts.

Keywords: Capital markets; Accounting choice; Earnings manipulation

Formulate Research Topics

Discussion:

- What is the research topic of Roychowdhury (2006)?

- Is it a good research topic?
  - Appropriateness: is it worthwhile? (clear link to theory, provide fresh insights, valuable findings)
  - Capability: is it feasible? (data availability, solid methodology, current)

Formulate Research Topics

Earnings Management:

Healy and Wahlen (1999): “Earnings management occurs when managers use judgment in financial reporting and in structuring transactions to alter financial reports to either mislead some stakeholders about the underlying economic performance of the company or to influence contractual outcomes that depend on reported accounting practices.”

Earnings = ACC + CFO

Earnings Management

- In accounting, earnings management is a method of manipulating financial records to improve the appearance of the company's financial position.
- Companies use earnings management to present the appearance of consistent profits and to smooth earnings' fluctuations.
- One of the most popular ways to manipulate financial records is to use an accounting policy that generates higher short-term earnings.

Examples: bad expense underprovision, delaying write-offs

Earnings = ACC + CFO

→ Accrual management (LIFO/FIFO) + real activity management (cut R&D)

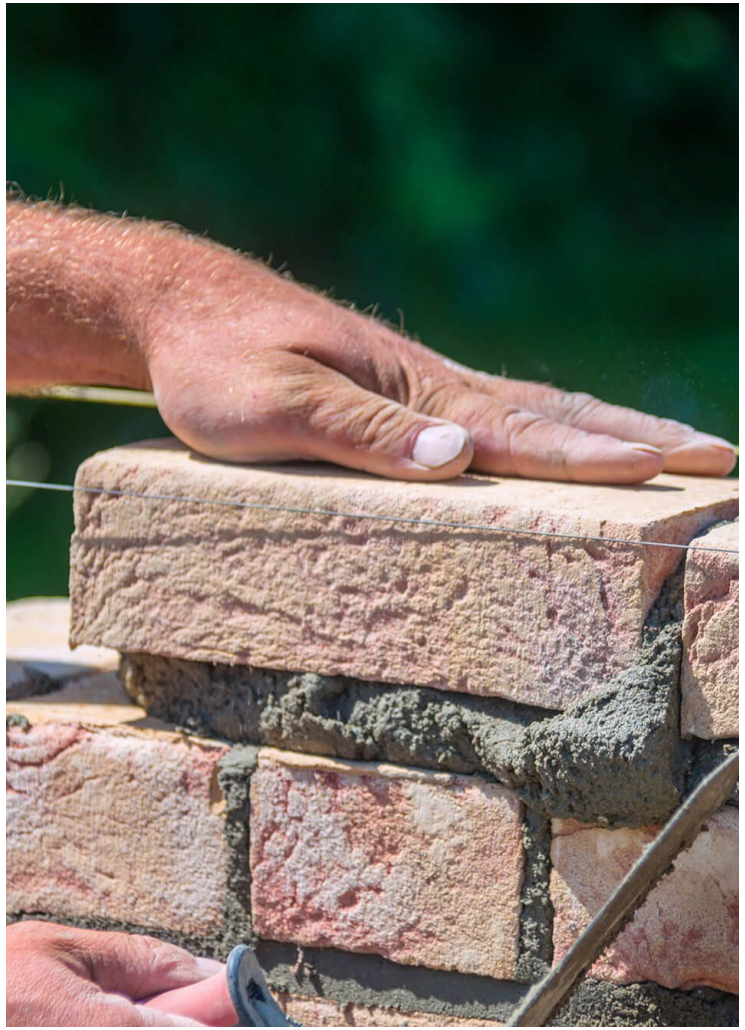
In-Class-Activity 1: Literature Review

- **Existing literature review on Earnings Management**
- **Literature review on Real Activities Manipulation**
- What's **the gap of the existing literature**? How to add a brick to the wall? What has not been explored/under explored?

In this part:

Identify your areas of literature

Identify the paragraph where gap is identified



In-Class-Activity 1: Literature Review

- **Earnings Management literature review:**
  - Healy (1985): The effect of bonus schemes on accounting decisions (JAE)
  - Defond and Jiambalvo (1994): Debt covenant violation and manipulation of accruals. (JAE)
  - Teoh et al (1998a): Earnings management and the long-run underperformance of seasoned equity offerings. (JFE)
  - Teoh et al (1998b): Earnings management and the long-run underperformance of initial public offerings. (JF)
  - Kasznik (1999): On the association between voluntary disclosure and earnings management (JAR)
  - Kothari (2001): Capital markets research in accounting (JAE)
  - Fields et al (2001): Empirical research on accounting choice (JAE)
  - Healy and Wahlen (1999): review of the earnings management literature and its implications for
  - standard setting, (AH)

In-Class-Activity 1: Literature Review

- **Gap in the existing literature:**

- Burgstahler and Dichev (1997) provide some limited evidence on whether executives manage real activities to meet the zero earnings threshold. They plot the 25th, 50th, and 75th percentiles of unscaled CFO for each earnings interval and find that the distribution of CFO shifts upward in the first interval to the right of zero. However, this preliminary evidence does not conclusively indicate real activities manipulation nor does it yield any insights into the activities underlying the patterns in CFO. Burgstahler and Dichev (1997) do not test whether the shifts are statistically significant, nor do they impose controls for firm size/performance.

In-Class-Activity 2: Hypotheses Development

- Read the paper and learn how to write the motivation of the hypotheses (reasons)
- Identify the hypotheses (there could be 2-5 hypotheses)

In-Class-Activity 2: Hypotheses Development

- Earnings = Sales – Expenses (COGS + other exps)
- Three real earnings manipulation methods:

1. **Sales manipulation**, that is, accelerating the timing of sales and/or generating additional unsustainable sales through increased price discounts or more lenient credit terms;
2. **Reduction of discretionary expenditures***; and
3. **Overproduction**, or increasing production to report lower COGS.

* Discretionary expenses are defined as the sum of (a) advertising expenses, (b) R&D expenses, and (c) selling, general and administrative (SG&A) expenses

In-Class-Activity 2: Hypotheses Development

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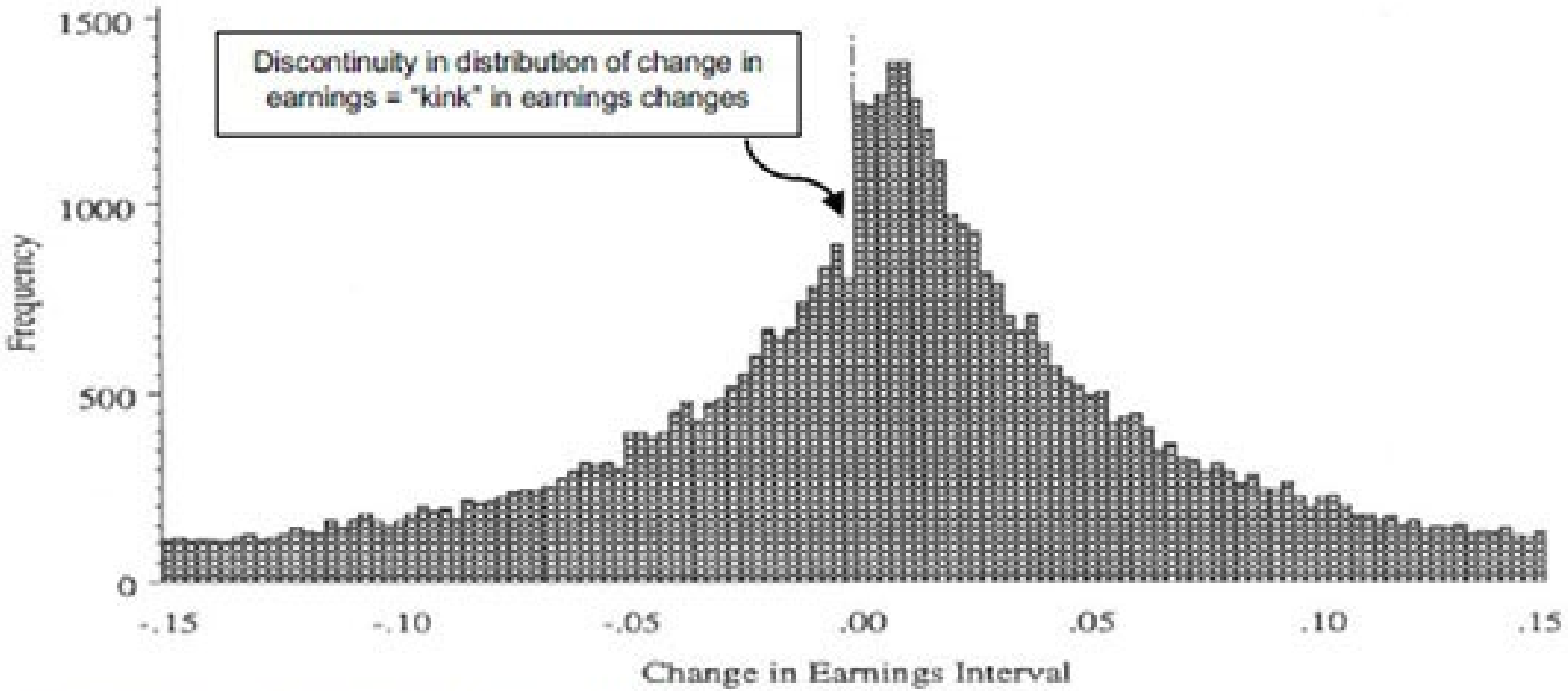
- **Main hypotheses:**
- Excessive price discounts and overproduction lead to abnormally high production costs relative to dollar sales. Reduction of discretionary expenditures leads to abnormally low discretionary expenses relative to sales;
- Price discounts, channel stuffing, and overproduction have a negative effect on contemporaneous abnormal CFO, while reduction of discretionary expenditures has a positive effect.
- ***H1A.*** *After controlling for sales levels, suspect firm-years exhibit at least one of the following: unusually low cash flow from operations (CFO) OR unusually low discretionary expenses.*
- ***H2A****. After controlling for sales levels, suspect firm-years exhibit unusually high production costs.*

Burgstahler and Dichev (1997)

Figure 2

The distribution of change in earnings of US non-financial companies. Change in earnings: Changes in annual net income scaled by market value of equity as at the beginning of the first year, ($Earnings_t$ – $Earnings_{t-1}$) / MV_{t-2}.



Source: Graph taken from Burgstahler and Dichev (1997b)

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- A) industry membership (manufacturing industry)
- B) incentives to meet zero earnings (debt, growth opportunities, short-term creditors)
- C) earnings management flexibility
- D) institutional ownership

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- **A) industry membership (manufacturing industry)**
- Industry membership: Overproduction and price discounts both generate abnormally high production costs relative to sales. Both manufacturing and non-manufacturing companies can offer price discounts to boost sales but overproduction as an earnings management strategy is only available to firms in manufacturing industries. Therefore, the evidence of abnormal production costs to be driven to a greater extent by suspect firm-years that belong to manufacturing industries.
- ***H3A.*** *Ceteris paribus, suspect firm-years in manufacturing industries exhibit higher abnormal production costs than other suspect firm-years.*

In-Class-Activity 2: Hypotheses Development

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Cross sectional variation in real activities manipulation:

- **B) incentives to meet zero earnings (debt)**
- Debt contracts include covenants that become tighter when firms incur losses. Losses would make these covenants more binding.

- ***H4A****. Ceteris paribus, suspect firm-years with debt outstanding exhibit abnormal production costs that are higher, and abnormal discretionary expenses that are lower than other suspect firm-years.*

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- **B) incentives to meet zero earnings (growth opportunities)**
- **Market-to-book:** Skinner and Sloan (2002) document that firms with growth opportunities are penalized more by the stock market when they miss earnings thresholds. While their study focuses primarily on analyst forecasts, it is likely that growth firms also experience pressure to meet other earnings thresholds, including zero.
- ***H5A****. Ceteris paribus, suspect firm-years with high market-to-book exhibit abnormal production costs that are higher than, and abnormal discretionary expenses that are lower than, other suspect firm-years.*

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- **B) incentives to meet zero earnings (short-term creditors)**
- **Short-term suppliers:** Graham et al. (2005) and Burgstahler and Dichev (1997): there are stakeholders of the firm who use heuristic cut-offs at zero to evaluate its performance. Among the stakeholders whom these studies identify are suppliers, lenders, employees, and customers worried about future services. If the firm's earnings performance falls below a certain threshold, like zero, the firm's ability to pay suppliers in time and its potential as a future buyer are in doubt. This leads suppliers to tighten terms of credit and other terms. Managers are more likely to worry about the negative reaction of suppliers if they have more trade credit and other short-term liabilities outstanding.
- ***H6A.*** *Ceteris paribus, suspect firm-years with high current liabilities as a percentage of total assets exhibit abnormal production costs that are higher than, and abnormal discretionary expenses that are lower than, other suspect firm-years.*

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- **C) Earnings management flexibility**
- Excess production to absorb fixed costs in inventory is easier to accomplish and more likely to escape detection when a firm traditionally maintains a high stock of inventory. Similarly, a firm with substantial credit sales to dealers can more easily engage in channel stuffing, or accelerating the recognition of sales by shipping goods early to its dealers and booking receivables.

- ***H7A.*** *Ceteris paribus, suspect firm-years with a high level of inventories and receivables as a percentage of total assets exhibit abnormal production costs that are higher than other suspect firm-years.*

In-Class-Activity 2: Hypotheses Development

Cross sectional variation in real activities manipulation:

- **C) institutional ownership**
- On the one hand, earning disappointments such as losses can possibly trigger institutional owners to engage in large-scale selling due to their focus on short-term earnings (myopic investment behavior). This implies that the presence of institutional shareholders exerts greater pressure on managers to avoid reporting losses. On the other hand, it is also possible that sophisticated institutional investors have a greater ability to analyze the long-term implications of current managerial actions. This would act as a disincentive for managers to engage in real activities manipulation, particularly if such manipulation reduces long-run firm value.

- ***H8A.*** *Ceteris paribus, suspect firm-years with high institutional ownership exhibit abnormal production costs that are lower, and abnormal discretionary expenses that are higher than other suspect firm-years.*

In-Class-Activity 3: Research Design

1. How to measure the key concept in the hypotheses
2. How to test the hypotheses
3. Any Limitations

Estimation models:

1. How to measure normal CFO?
2. How to measure normal Production cost (PROD)?
3. How to measure normal Discretionary Expense?

In-Class-Activity 3: Research Design

$$\mathrm{CFO}_t/A_{t-1} = \alpha_0 + \alpha_1(1/A_{t-1}) + \beta_1(S_t/A_{t-1}) + \beta_2(\Delta S_t/A_{t-1}) + \varepsilon_t, \quad (1)$$

$$\mathrm{COGS}_t/A_{t-1} = \alpha_0 + \alpha_1(1/A_{t-1}) + \beta(S_t/A_{t-1}) + \varepsilon_t. \quad (2)$$

$$\Delta\mathrm{INV}_t/A_{t-1} = \alpha_0 + \alpha_1(1/A_{t-1}) + \beta_1(\Delta S_t/A_{t-1}) + \beta_2(\Delta S_{t-1}/A_{t-1}) + \varepsilon_t, \quad (3)$$

$$\mathrm{PROD}_t/A_{t-1} = \alpha_0 + \alpha_1(1/A_{t-1}) + \beta_1(S_t/A_{t-1}) + \beta_2(\Delta S_t/A_{t-1}) + \beta_3(\Delta S_{t-1}/A_{t-1}) + \varepsilon_t. \quad (4)$$

$$\mathrm{DISEXP}_t/A_{t-1} = \alpha_0 + \alpha_1(1/A_{t-1}) + \beta(S_{t-1}/A_{t-1}) + \varepsilon_t. \quad (5)$$

In-Class-Activity 3: Research Design

How to select suspect firm-years?

How to increase the power of tests to detect real activities manipulation?

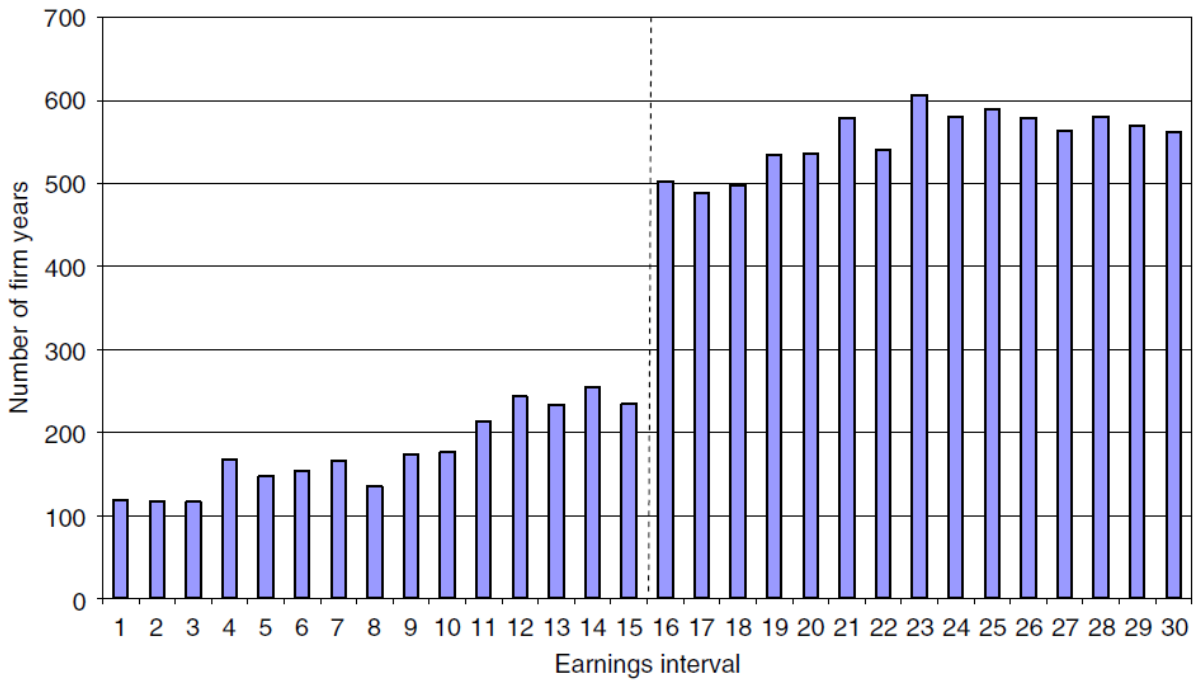

Fig. 1. Number of firm years by earnings interval: 21,758 firm-years over the period 1987–2001 are classified into earnings intervals over the range −0.075 to +0.075, where earnings is defined as income before extraordinary items (IBEI) scaled by total assets (A). Each interval is of width 0.005, with category 16 including firm-years with earnings greater than or equal to zero and less than 0.005. The figure is truncated at the two ends and includes 10,958 firm-years.

In-Class-Activity 3: Research Design

- How to compare suspect firm-years with the rest of the sample?

$$Y_t = \alpha + \beta_1(\text{SIZE})_{t-1} + \beta_2(\text{MTB})_{t-1} + \beta_3(\text{Net income})t + \beta_4(\text{SUSPECT_NI})_t + \varepsilon_t.$$

In-Class-Activity 3: Research Design

- How to select suspect firm-years?
- How to increase the power of tests to detect real activities manipulation?

$$\text{Abnormal CFO}_t = \alpha + \beta_1(\text{SIZE})_{t-1} + \beta_2(\text{MTB})_{t-1} + \beta_3(\text{Net income})_t + \varepsilon_t.$$

$$\text{Abnormal PROD}_t = \alpha + \beta_1(\text{SIZE})_{t-1} + \beta_2(\text{MTB})_{t-1} + \beta_3(\text{Net income})_t + \varepsilon_t.$$

$$\text{Abnormal DISEXP}_t = \alpha + \beta_1(\text{SIZE})_{t-1} + \beta_2(\text{MTB})_{t-1} + \beta_3(\text{Net income})_t + \varepsilon_t.$$